

Dr. Ilaria Tabone

Date of birth: 25/08/1988

Civil state: single

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Professional employment

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| 02.08.2024 - ongoing | Assistant Professor , Department of Geophysics, University of Concepción, Concepción, Chile. |
| 01.11.2022-30.04.2024 | Post-doctoral researcher (temporal position for 3 years with third-party funding - DFG), Institute of Geography, FAU, Erlangen, Germany. |
| 01.07.2020-31.10.2022 | Post-doctoral researcher (AvH Fellow), Institute of Geography, University of Erlangen-Nuremberg (FAU), Erlangen, Germany. |
| 16.03.2019-30.06.2020 | Post-doctoral researcher, Complutense University of Madrid (UCM), Madrid, Spain. |
| 04.02.2019-03.02.2020 | Post-doctoral researcher, Complutense University of Madrid (UCM), Madrid, Spain. |

Academic education

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| 11.01.2016-10.01.2019 | Doctorate in Physics, Complutense University of Madrid (UCM), Madrid, Spain. Thesis: <i>The role of ice-ocean interactions in the past evolution of the Greenland Ice Sheet</i> . Supervisors: Prof. Dr. Marisa Montoya and Dr. Alexander Robinson. |
| 09.2011-15.04.2015 | Master in Physics of Environment and Advanced Technologies, University of Torino, Torino, Italy. Thesis: <i>Comparison of meteorological observations and WRF model retrieved Cloud Top Height for the JEM-EUSO experiment</i> . Grade: 110/110 cum Laude. |
| 09.2007-08.04.2011 | Bachelor in Physics, University of Torino, Torino, Italy. Grade: 108/110. |

Personal grants and awards

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| 2025 | Fondecyt Iniciación 2025 . Project title: "Projecting the Patagonian Icefields mass loss in the future (PROPAGEMOS)." Principal Investigator. N. 11251090. From 01.04.2025 to 31.03.2028. 30.000 euros /year |
| 2022 | DFG Individual Research Grant . Project title: " <i>Regional-scale ice-flow modelling in Patagonia since the Little Ice Age (RESPONSE)</i> ". Host institution: Institute of Geography, FAU, Germany. Project n. 495516510. Principal Investigator. From 01.10.2022 to 31.09.2025. 88447 euros/year. |
| 2020 | Alexander Von Humboldt (AvH) Research Fellowship for Postdoctoral Researchers (grant 1212941-ITA-HFST-P). Project title: " <i>Improving surface mass balance and hydrology representation at the Northeast Greenland (SURFYNG)</i> ". Host institution: Institute of Geography, FAU, Germany. Fellowship from 01.07.2020 to 31.10.2022. |
| 2019 | Post-doctoral fellowship from the Spanish National Programme for the Promotion of Talent and Its Employability (1 year). |
| 2017 | Early Career Scientist's Travel Grant, EGU, General Assembly 2017. |

- 2016 **Pre-doctoral fellowship** from the Spanish National Programme for the Promotion of Talent and Its Employability (3 years).
- 2015 ERASMUS+ Traineeship grant (4 months). Universidad Carlos III, Madrid (Spain).

Research interests

Ice-flow modelling, ice-ocean, ice-atmosphere and ice-solid earth interactions, glacier hydrology, paleoclimatology, Greenland and Antarctic Ice Sheets, Patagonian ice fields, climate change.

Publications

1. Gutiérrez-González, L., Robinson, A., Alvarez-Solas, J., **Tabone, I.**, Swierczek-Jereczek, J., Moreno-Parada, D., and Montoya, M.: Hysteresis of the Greenland ice sheet from the Last Glacial Maximum to the future, *The Cryosphere*, 20, 1139-1162, <https://doi.org/10.5194/tc-20-1139-2026>, 2026.
2. Farías-Barahona D, Braun MH, Schaefer M, ..., **Tabone I.**, QFuego-Patagonia: a comprehensive glacier-related dataset for Patagonia and Tierra del Fuego, South America. *Journal of Glaciology*. 2026;72:e14. doi:10.1017/jog.2025.10110, 2025.
3. Herrmann O., Groos A., **Tabone I.**, Juvet G., Fuerst J., "A Kalman Filter-based Framework for Assimilating Remote Sensing Observations into a Surface Mass Balance Model", *Annals of Glaciology* 66, e23, 1-14. <https://doi.org/10.1017/aog.2025.10020>
4. Lutz, K., **Tabone, I.**, Humbert, A., and Braun, M.: Multi-annual patterns of rapidly draining supraglacial lakes in Northeast Greenland, *The Cryosphere*, 19, 2601–2614, <https://doi.org/10.5194/tc-19-2601-2025>, 2025.
5. Blasco J., **Tabone I.**, Moreno-Parada D., Robinson A., Álvarez-Solas, J., Pattyn F., Montoya M. (2024): Antarctic Tipping points triggered by the mid-Pliocene warm climate, *Climate of the Past*, 20, 1919–1938, <https://doi.org/10.5194/cp-20-1919-2024>.
6. **Tabone, I.**, Robinson A., Montoya M., Álvarez-Solas, J.: Holocene thinning in central Greenland controlled by the Northeast Greenland Ice Stream, *Nat Commun*, 15, 6434 (2024). <https://doi.org/10.1038/s41467-024-50772-5>.
7. Adams Jr, J. H., Ahmad, S., Allard, D., Anzalone, A., Bacholle, S., Barrillon, P., Bayer, J., Bertaina, M., Bisconti, F., Blaksley, C., Blin-Bondil, S., Bobík, P., Cafagna, F., Campana, D., Capel, F., Casolino, M., Cassardo, C., Catalano, C., Cremonini R., Dagoret-Campagne, S., Danto, P., del Peral, L., de la Taille, C., Díaz Damian, A., Dupieux, M., Ebersoldt, A., Ebisuzaki, T., Eser, J., Evrard, J., Fenu F., Ferrarese S., Fornaro, C., Fouka, M., Gorodetzky, P., Guarino, F., Guzman, A., Hachisu Y., Haungs, A., Judd, E., Jung, A., Karczmarczyk, J., Kawasaki, Y., Klimov, P.A., Kuznetsov, E., Mackovjak, S., Manfrin, M., Marcelli, L., Medina-Tanco, G., Mercier, K., Merino, A., Mernik, T., Miyamoto, H., Morales de los Ríos, J.A., Moretto, C., Mot, B., Neronov, A., Ohmori, H., Olinto, A.V., Osteria, G., Panico, B., Parizot, E., Paul, T., Picozza, P., Piotrowski, L.W., Plebaniak, Z., Pliego, S., Prat, P., Prévôt, G., Prieto, H., Putis, M., Rabanal, J., Ricci, M., Rojas, J., Rodríguez Frías, M.D., Roudil, G., Sáez Cano, G., Sahnoun, Z., Sakaki, N., Sanchez, J.C., Santangelo, A., Sarazin, F., Scotti, V., Shinozaki, K., Silva, H., Soriano, J.F., Suino, G., Szabelski, Toscano, S., **Tabone I.**, Takizawa, Y., von Ballmoos, P., Wiencke, L., Wille, M., and Zotov, M. (2022). A Review of the EUSO-Balloon Pathfinder for the JEM-EUSO Program. *Space Science Reviews*, 218(1), 3.
8. **Tabone, I.**, Robinson, A., Alvarez-Solas, J., Montoya, M. (2019): Submarine melt as a potential trigger for the North East Greenland Ice Stream margin retreat during Marine-Isotope Stage 3, *The Cryosphere*, 13, 1911-1923, doi: 10.5194/tc-13-1911-2019.
9. **Tabone, I.**, Robinson, A., Alvarez-Solas, J., Montoya, M. (2019): Impact of millennial-scale oceanic variability on the Greenland ice-sheet evolution throughout the last glacial period, *Climate of the Past*. 15, 593-609, doi: 10.5194/cp-15-593-2019.

10. Blasco, J., **Tabone, I.**, Robinson, A., Álvarez-Solas, J. and Montoya, M. (2019): The Antarctic Ice Sheet response to glacial millennial-scale variability, *Climate of the Past*, 15, 121-133, doi: 10.5194/cp-15-121-2019.
11. Anzalone, A., Bertaina, M. E., Briz, S., Cassardo, C., Cremonini, R., de Castro, A. J., Ferrarese, S., Isgró, F., Lopez F., **Tabone, I.** (2018): Methods to Retrieve the Cloud-Top Height in the Frame of the JEM-EUSO Mission. *IEEE Transactions on Geoscience and Remote Sensing*, 99, 1-15, doi: 10.1109/TGRS.2018.2854296.
12. **Tabone, I.**, Blasco, J., Robinson, A., Alvarez-Solas, J., Montoya, M. (2018): The sensitivity of the Greenland Ice Sheet to glacial-interglacial oceanic forcing, *Climate of the Past*, 14, 455-472, doi: 10.5194/cp-14-455-2018.
13. **Tabone, I.**, Briz, S., Anzalone, A., De Castro, AJ, Lopez, F., Ferrarese, S., Isgró, F., Cassardo, C., Cremonini, R., Bertaina, M., Comparing different methods to retrieve cloud top height from Meteosat satellite data, *Proc. SPIE 9640, Remote Sensing of Clouds and the Atmosphere XX*, 96400C (16 October 2015); <https://doi.org/10.1117/12.2195012>.

Book Chapters

Álvarez-Solas, J., Blasco, J., Gabarró, C., Montoya, M., Robinson, A., **Tabone, I.** (2021), *Criosfera: el hielo polar y su papel en el clima terrestre*, chapter of *Observando los polos: Una visión integral del conocimiento de las zonas polares y sobre las alteraciones que están sufriendo debido al cambio global* (2021) ,Versión del editor: <http://editorial.csic.es/publicaciones/libros/13561/0/observando-los-polos.html> URI: <http://hdl.handle.net/10261/261503> ISBN: 978-84-00-10865-6.

Stays abroad

- 2026. Visiting scientist UCM, España, 18-20 May 2026
Visiting scientist FAU Alemania, 4-8 May 2026
- 2025. Visiting scientist, FAU, Alemania, 7-14/07/2025
- 2023 Visiting scientist, Universidad Austral de Chile (UACH), Chile, 24 Oct-17 Dic 2023.
Visiting scientist, Universidad Austral de Chile (UACH), Chile, 28 Mar-13 Apr 2023.
Visiting scientist, FAUG, Universidad de Concepción (Udec), Chile, 24-27 Mar 2023.
- 2019 Visiting scientist, Institute of Geography, University of Erlangen-Nuremberg, (FAU), Erlangen, Germany, May-Oct. 2019.
- 2015 Erasmus+ traineeship, Carlos III University, Madrid (Spain), May-Aug 2015

Participation in workshops, summer schools and courses

2026 “Taller lidera mujeres en la academia” enero 2026

- 2023 *Greenland Ice Sheet Stability* Workshop, University of Bergen, Norway, 19-21 Apr 2023.
Northeast Greenland Workshop, Aarhus, Denmark, 25-27 Jan 2023.
- 2020 *Beginner Elmer/Ice* course, online, 16-27 Nov 2020.
- 2018 *Earth system variability through time*, 17th International Swiss Climate Summer School, Grindelwald (Switzerland), 26-31 August 2018.
- 2017 *Ice and Climate*, Karthaus Summer School, Karthaus (Italy), Sept. 2017.
- 2016 *Glaciers, Ocean, Climate and Paleoclimate*, Course for PhD students (IMEIO-UPM) *CESM Tutorial*, NCAR, Boulder (CO), USA, 8-12 Aug. 2016.

Paleoclimate, glacier surface mass balance and coupled climate-ice sheet models, Course for PhD students (IMEIO-UPM), Mar. 2016.

Modeling and Glacial Dynamics, Course for PhD students (IMEIO-UPM), Jan 2016.

Contributions to conferences (selected)

VII congreso de SOCHICRI, poster, Coyhaique (Chile), "Detection and analysis of one GLOF event at Almena Glacier, Cordillera Darwin, via satellite monitoring and Ferrybox data", Ilaria Tabone, Schliermann, Giesecke, Arndt, Temme, Fonseca Gallardo, Narvaez, Fairas-Barahona, 8-10/10 2025.

SCAR XI Open Science Conference, *oral presentation*, Pucón (Chile), 08/2024. "Modelling the Southern Patagonia Icefield at the end of the Little Ice Age.", Ilaria Tabone, Johannes Fürst, Marius Schaefer and Matthias Braun. 19/08/2024-23/08/2024

European Geoscience Union (EGU), General Assembly 2024, *poster*, Vienna (Austria). "Detection and analysis of GLOF events at Alemanía (Roncagli) Glacier, Cordillera Darwin". Maria Schliermann, Ilaria Tabone, David Fariás-Barahona, Jan Erik Arndt, Ricardo Giesecke. 14-19 April 2024

European Geoscience Union (EGU), General Assembly 2023, *poster*, Vienna (Austria). "Seasonal and inter annual effects of meltwater runoff on ice dynamics in Northeast Greenland". Tabone, fürst, Mölg, 24-28/04/2023

American Geoscience Union (AGU), General Assembly 2022, *invited talk*, Chicago (USA). "On the stability of Northeast Greenland: an interplay between surface processes, basal hydrology and ice dynamics". Tabone, Fuerst, Mölg., 12-16/12/2022

European Geoscience Union (EGU), General Assembly 2022, *oral presentation*, Vienna (Austria). "Investigating the influence of increased meltwater runoff on basal sliding in Northeast Greenland". Tabone, Fuerst, Mölg., 23-27/05/2022

28th International Polar Conference, 01-05 May 2022, *poster*, Potsdam (Germany). "Modelling surface meltwater and ice dynamics interactions at the Northeast Greenland with a fully-coupled ice-flow - hydrology model". Tabone, Fuerst, Mölg.

European Geoscience Union (EGU), General Assembly 2021, *oral presentation*, online event. "Modelled Holocene thinning in Greenland improved by transient past climatologies and fast flow". Tabone I., Robinson A., Alvarez-solas J., Blasco J., Moreno D., Montoya M. , 19-30/04/2021

European Geoscience Union (EGU), General Assembly 2020, *oral presentation*, online event. "A multi-approach skill-score procedure to optimize continental-scale ice-sheet models", Tabone I., Robinson A., Alvarez-solas J., Blasco J., Moreno D., Montoya M., 04-08/05/2020

European Geoscience Union (EGU), General Assembly 2019, *oral presentation*, Vienna (Austria). "Submarine melt as a potential trigger of the North East Greenland Ice Stream margin retreat during MIS-3". Ilaria Tabone (1,2), Alexander Robinson (1,2), Jorge Alvarez-Solas (1,2), Marisa Montoya. 07-12/04/2019

European Geoscience Union (EGU), General Assembly 2018, *two posters*, Vienna (Austria). "Optimizing ice-sheet-shelf model performance through a skill-score study" Ilaria Tabone, Javier Blasco, Alexander Robinson, Jorge Alvarez-Solas, Marisa Montoya. "The Greenland Ice Sheet sensitivity to oceanic forcing on a glacial-interglacial timescale", Ilaria Tabone, Javier Blasco, Alexander Robinson, Jorge Alvarez-Solas, Marisa Montoya. 04-08/05/2018

5th PAGES Open Science Meeting, 7-9 May 2017, *oral presentation*, Zaragoza (Spain). "Sensitivity of the Greenland Ice Sheet to oceanic changes in the last 150 kyrs", Ilaria Tabone, Alexander Robinson, Jorge Alvarez-Solas, Marisa Montoya., 23-28/04/2017

European Geoscience Union (EGU), General Assembly 2017, *oral presentation*, Vienna (Austria). "The Greenland Ice Sheet-ocean interaction in the past two glacial cycles", Ilaria Tabone, Alexander Robinson, Jorge Alvarez-Solas, Marisa Montoya.,

XI Congress TOPCART 2016, *oral presentation*, Toledo (Spain), 26-20/10/2016.

Peer reviewing

Nature Geoscience, The Cryosphere, Journal of Glaciology, Climate of the Past, Geophysics, Paleoceanography and Paleoclimatology, PLOS One.

Teaching experience: courses and seminars

“Métodos Numéricos con Python” for the Bachelor Degree in Geophysics/Meteorology and for Master Degree in Geophysics at the University of Concepción (I semester 2025).

“Global physical climatology” for the Bachelor degree in Geophysics/Meteorology at the University of Concepción (II semester 2024).

Seminar “La física de los hielos continentales y su respuesta al cambio climático” for undergraduate students in Physics (Uach, Dec. 2023).

Seminars series at the summer school in Geophysics for undergraduate and postgraduate students of the 'Physics Advanced' program within the framework of the 'Elite Graduate Programme' at FAU (Sept. 2023). Seminar titles: “Introduction to Glacier Physics - Part I and Part II” and “Response of Ice Bodies to Climate Change”. Invited speaker.

Seminar “Evolución pasada y futura de los mantos de hielo polares y campos de hielo patagónicos simulada por modelos numéricos” for undergraduate and postgraduate students (Udec, April 2023, in Spanish). This same talk was also presented at Uach (April 2023) and at UCM (Spain, Dec. 2023), as part of the seminar series for the Master's program in Meteorology and Geophysics.

Seminar “Ice Dynamics - Mechanisms and Modeling” for postgraduate students, M3OCCA doctoral program (FAU, Nov. 2022).

Seminar “Groenlandia” for postgraduate students of the Magister en Recursos Hídricos (Uach, Nov. 2021, online).

Teaching assistant for the course “Introduction to Geophysical modelling” within the Bachelor degree program in Physical Geography, FAU (2022/2023).

Teaching assistant for the course “Laboratory of Physics applied to Biology” within the Bachelor degree program in Biology, UCM (2018/2019).

Teaching assistant for the course “Physics II” within the Bachelor degree program in Geoscience Engineering, UCM (2017/2018).

Participation in tribunals

Committee member in the PhD defense tribunal of Kaian Fernandes Shahateet at the Politecnical University of Madrid (2023), within the programme of Mathematical Engineering, Statistics, and Operations Research. Also served as Thesis reviewer.

Committee member in the appointment process for the Full Professor position in Cultural Geography (FAU), representing the mid-level academic staff (2022-2023).

Commissions

03/26 to ? Comision de Convivencia Genero, inclusion de la facultad de CFM

Since June 2026 IGS retention and new members commission

Supervision

Ongoing Paul Sandoval PhD in Environmental Engineering

Raul Correa MSc in Geophysics
Mañio Zúñiga MSc in Geophysics
Bastian Espinoza MSc in Geophysics

- Ongoing Maria Schliermann's Master thesis in Oceanography (UdeC), titled "Impacto del océano en el derretimiento submarino de glaciares en sistemas de fiordos de Patagonia y Tierra del Fuego", *ongoing*.
- 2024 Maria Schliermann's Bachelor thesis in Physical Geography (FAU), titled "Water-Glacier Interaction in Cordillera Darwin - Water mass characterisation and GLOFs".
- 2022 Jacob Alberto García Suárez's Master Thesis in Meteorology and Geophysics (UCM, Spain), titled "The Evolution of the Greenland Ice Sheet under Global Warming".

Fieldwork

2026. Fieldwork at glacier HPN4, 2-5 Abril 2026

2026 Campaña oceanografica Patagonia Norte Instituto de Fomento Pesquero (IFOP) bathymetry measurements, CTD casts. 14-25 enero 2026.

2025/2026 Monitoreo Complejo volcánico de Chillan. (Desde octubre a marzo 2026 par ale balance de masa anual del glaciar)

2025 Rock glacier campaign in the Nevado Valley, Chile, 6-9 March (geophysical measurements).

2024 Pio XI glacier campaign (Southern Patagonian Icefield), Chile, 01-10 Nov (exploration, ablation stake installation and air temperature sensor installation).

2023 Glaciar Mocho-Choschuenco (Late winter Surface Mass Balance), Chile, 3-4 Dic (ablation stakes measurements and snow pits).

Kanderfirn Gletscher (Summer SMB), Kanderfirn, Swiss Alps, repeated visits in Summer for annual glaciological mass balance.

Kanderfirn Gletscher (Winter SMB), Kanderfirn, Swiss Alps, 26-28 May for annual glaciological mass balance.

Exploration in Cordillera Darwin: Glaciers Marinelli, Renland, Videla, Chile, 13-17 March.

2022 Kanderfirn Gletscher (Summer SMB), Kanderfirn, Swiss Alps, 4-9 Sept 2023 for annual glaciological mass balance.

Outreach

Invited speaker for "Webinar conversatorio 1er día nacional por la conservación de los glaciares", organized by Plantae Fundación and UNESCO. 27/3/26

Columna de Opinion diario de chillan.

Participant BBC documentary (Mysterious Earth).

Participant in Diálogos Contemporáneos UdeC, 28 Mayo 2025, Casa del Arte, UdeC "Glaciares ¿son importantes para nuestro pais?", transmitido por TVU.

Pais circular

Organizador de charlas "Sin hielo no hay frappé", en el marco del "Día de glaciares", Biblioteca Municipal de Concepción, 21 Marzo 2025.

Participant in the Podcast EpiSTEMas, Feb 2025.

Ladera sur, Pio XI, 2024.

Participant in the "Escuela de Verano en Geofisica", Universidad de Concepción, 07-09 Jan 2025.

Participant and organiser of the outreach event "Lange Nacht der Wissenschaften" 21/10/2023.

Contributor to the EGU Blog Cryosphere.

Contributor to the fact-checking website "Maldita.es"

Asesoria

DGA Monitoreo complejo volcánico de chillan

DIFROL asesoria de modelación glaciologica???

Languages

Italian	Fluent (Mothertongue)
Spanish, English	Fluent
German	Proficient
French	Basic

Concepción, 01/06/2026